A four-view smartphone video dataset of a robot-held flexible strip, with camera poses and a 3D LaGSplat baseline

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October 2026

arXiv:2611.09959v1 [cs.CV] 26 Oct 2026

This note describes a multi-view video dataset of a deformable object driven by a measured input, with the measured pose of every camera, and a first 3D result obtained on it with LaGSplat [1]. Please cite this note when using the dataset. LaGSplat learns, from video, a low-dimensional latent state $\mathbf{q}$ that plays two roles at once: it is the generalized coordinate of a learned dissipative Lagrangian, and the conditioning variable of a Gaussian Splatting decoder whose primitives are explicit points $\mu_i(\mathbf{q})$ that move with the object. A force $\mathbf{f}$ applied on the reconstructed object then enters the equations of motion as the generalized force $J(\mathbf{q})^\top \mathbf{f}$, with $J = \partial\mu/\partial\mathbf{q}$. Fitting the decoder in 3D calls for synchronized multi-view footage. In [1], the 3D experiments use synthetic views rendered by SHARP; this dataset and the results below are not part of it.

Setup. A robot arm holds a flexible strip, about 23 cm long, in a gripper. The end effector moves by jerks along a nearly one-dimensional horizontal path; after each jerk the strip oscillates freely and its motion decays. Four iPhones film the scene at 1920×1080 and 60000/1001 fps. Three of them (iphone0, iphone1, iphone2) are fixed and mounted rigidly to one another; iphone0 is a different model from the other three. The fourth, iphone3, is handheld and moves around the scene for the first seven minutes, then is held nearly still. In a separate session, with every object in the room still, robot included, iphone3 filmed the same room in two clips, which give a static reconstruction of the room (Fig. 1).

Synchronization. The coarse offsets come from a clap and from the motion of the arm, whose jerks form an irregular, hence unambiguous, sequence of instants. The clock rates, read on the container timestamps, are not equal: over the recording, iphone0 drifts by 9 frames with respect to the three others. All four videos are therefore resampled on the clock of iphone1: frame k of every video shows the same instant, taken as the nearest source frame, with a residual of at most half a frame (8 ms). Each video has 36 734 frames (612.8 s).

Camera geometry. No calibration target was filmed; all poses come from the scene. The world frame is the camera frame of iphone0 (OpenCV convention), and the unit of length is the distance between iphone0 and iphone1 (about 0.38 m).

- Fixed cameras: focal lengths, radial distortion and poses are solved by a joint bundle adjustment on manually clicked correspondences. To check the result, points are triangulated from iphone0 and iphone1 only, then projected into iphone2: they fall 3 to 6 px from where they appear in its image.
- Handheld camera: one pose per frame, by PnP against a SIFT map built from iphone3's own frames. Poses with too few inliers or with non-physical jumps are removed: 25 381 of the 36 734 frames have a pose.
- Static clips: structure from motion with COLMAP on key frames, aligned to the world frame by a similarity. A block of poses had been placed on the wrong side of a nearly symmetric cabinet; these are corrected and flagged.

State and zones. The motion of the arm has a single parameter. The file `state.csv` gives it per frame as $s(t)$, the arc length of the gripper along its path, tracked in iphone1; it is the measured input of the system. The same file gives the latent state $\mathbf{q}(t) \in \mathbb{R}^2$ of the strip, as encoded by the baseline below, for reference. The file `zones.json` gives two oriented boxes in the world frame (Fig. 1): the zone q contains the strip, the zone s the moving part of the arm.

Baseline. We fit LaGSplat in 3D on this dataset, with three changes to [1]. (i) The arm is an input: the decoder is conditioned on $(s, \mathbf{q})$ and the dynamics is $M\ddot{\mathbf{q}} = -\nabla V(\mathbf{q}) - C\dot{\mathbf{q}} + G\ddot{s}$, with constant mass M and dissipation C and an invex potential V, which has a single minimum. The acceleration of the gripper enters as a forcing term, in the same way as the pressure forcing of [1]. (ii) The affine decoder is replaced by a shared deformation network that outputs the displacement, scale, rotation and colour of each Gaussian. J is no longer constant and is computed at the current state; the force transport $J^\top \mathbf{f}$ is unchanged, since it needs explicit moving primitives, not a linear decoder. (iii) The background is a static reconstruction of the room from the static clips; only the Gaussians in the zone q depend on $(s, \mathbf{q})$, those in the zone s on s alone. The model has about 10^5 Gaussians,

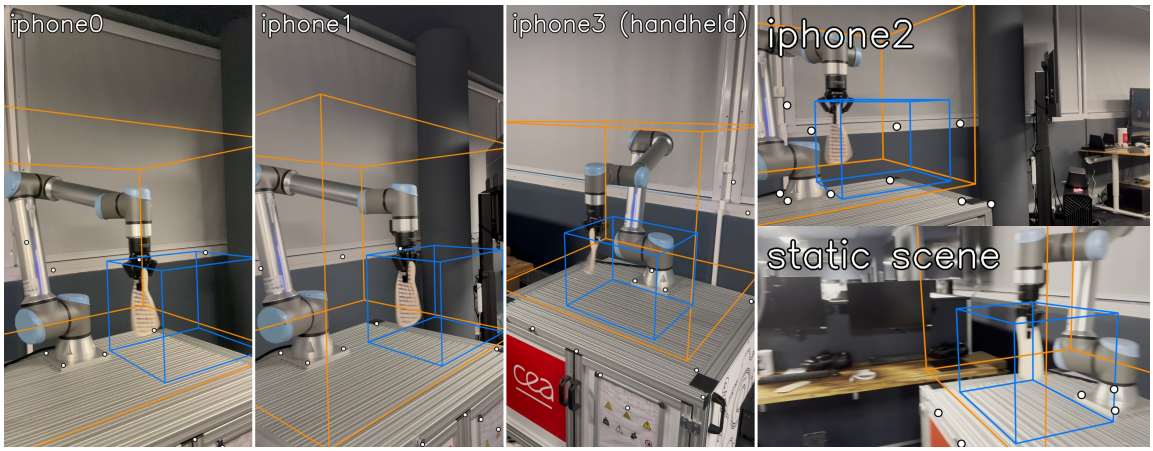


Figure 1: The four synchronized views at the same frame k, and one frame of the static scene. The zone s (orange), the zone q (blue) and the static landmarks (white) are projected with the measured pose of each camera.

File	Content
`scene_mobile/iphone{0,1,2,3}.mp4`	synchronized session, H.264, 36 734 frames each, no audio
`scene_mobile/frames.csv`	per frame k: common time, source frame of each camera, residual offset
`scene_mobile/sync.json`	cut, clock rates, dropped frames, checks
`scene_statique/iphone3_fixed_scene{,_2}.mp4`	static room, 1920 × 1080, 30 fps, 6 303 and 1 722 frames
`geometry/cameras_fixed.json`	intrinsics, distortion and pose of iphone0, 1, 2
`geometry/iphone3_poses.csv`	pose of iphone3 per frame k (with `iphone3_intrinsics.json`)
`geometry/static_poses.json`	poses of the static clips, in the same world frame
`geometry/landmarks.json`, `zones.json`	3D static landmarks; zones s and q
`geometry/state.csv`	per frame k: input $s(t)$ and baseline latent state $\mathbf{q}(t)$

Table 1: Files of the dataset.

and the camera poses are refined during training. On the last 60 s, held out, the rendered images (960 px) reach a PSNR of 31.4 to 32.8 dB for the fixed cameras and 26.5 dB for the handheld one; restricted to the projection of the zone q, which contains the strip, 28.4 to 30.0 dB and 26.2 dB. Held-out images of the static scene reach 24.7 dB, after a gain and bias per colour channel. For the dynamics, rollouts are started from the measured state on the held-out part; their RMS error, divided by the standard deviation of $\mathbf{q}$, is 0.53 over 0.25 s and 0.83 over 2 s. The model runs as a real-time 3D demo in a web browser, in which the strip responds to forces applied with the mouse: `https://louenpottier.github.io/xrdays.html`.

Limitations. The calibration error is a few pixels, not sub-pixel, and some poses of iphone3 are wrong during the handheld part. $s(t)$ is read in one view, in pixels. The colours of iphone0 differ from the other cameras.

Availability. Dataset and checkpoints: `10.5281/zenodo.23601189`, licence CC BY-NC 4.0.
Training code: `https://github.com/LouenPottier/lagsplat-mpov`.

References

[1] Louen Pottier. LaGSplat: Inferring physics-governed interactive simulation from monocular video using latent lagrangian Gaussian splatting. arXiv preprint arXiv:2608.16324, 2026.